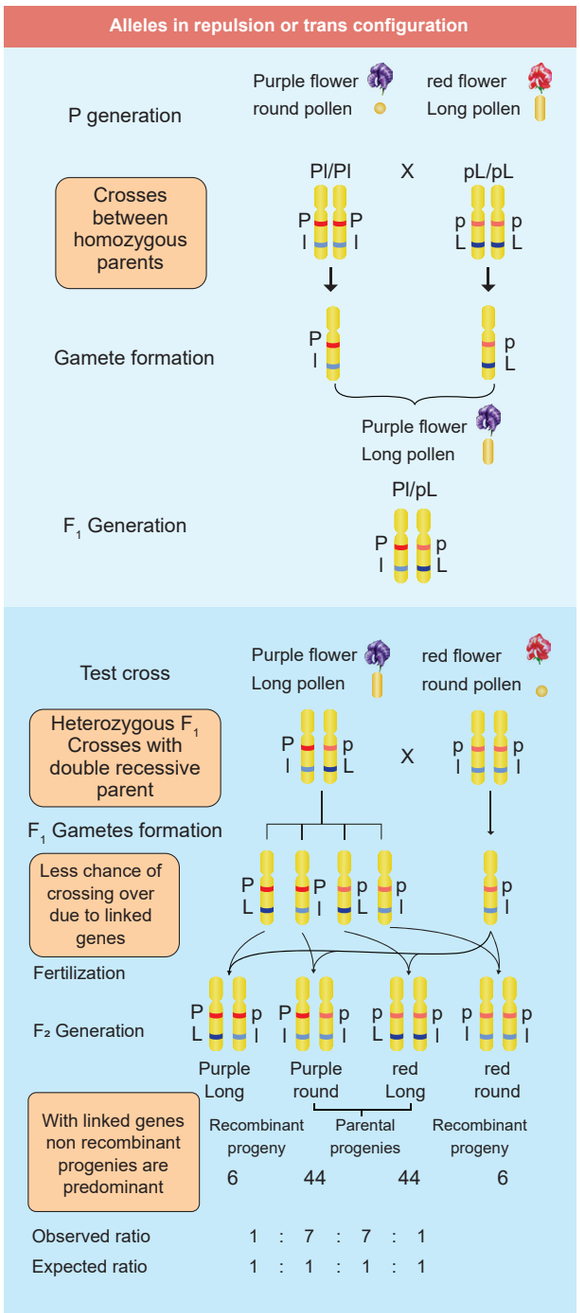



**Figure 3.5:** Alleles in repulsion or trans configuration

| Name of organism | Linkage groups |
|---|---|
| *Mucor* | 2 |
| *Drosophila* | 4 |
| Sweet pea | 7 |
| *Neurospora* | 7 |
| Maize | 10 |

**Table 3.3 :** Linkage groups in some organisms

Linkage and crossing over are two processes that have opposite effects. Linkage keeps particular genes together but crossing over mixes them. The differences are given below.

| Linkage | Crossing over |
|---|---|
| 1. The genes present on chromosome stay close together | It leads to separation of linked genes |
| 2. It involves same chromosome of homologous chromosome | It involves exchange of segments between non-sister chromatids of homologous chromosome. |
| 3. It reduces new gene combinations | It increases variability by forming new gene combinations. lead to formation of new organism |

**Table 3.4:** Differences between linkage and crossing over

## 3.3 Crossing Over

Crossing over is a biological process that produces new combination of genes by inter-changing the corresponding segments between non-sister chromatids of homologous pair of chromosomes. The term 'crossing over' was coined by **Morgan (1912).** It takes place during pachytene stage of prophase I of meiosis. Usually crossing over occurs in germinal cells during gametogenesis. It is called meiotic or germinal crossing over. It has universal occurrence and has great significance. Rarely, crossing over occurs in somatic cells during mitosis. It is called somatic or mitotic crossing over.

### 3.3.1 Mechanism of Crossing Over

Crossing over is a precise process that includes stages like synapsis, tetrad formation, cross over and terminalization.

#### (i) Synapsis

Intimate pairing between two homologous chromosomes is initiated during zygotene stage of prophase I of meiosis I. Homologous chromosomes are aligned side by side resulting in a pair of homologous chromosomes called **bivalents**. This pairing phenomenon is called **synapsis or syndesis.** It is of three types,



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